Math 125 - Winter 2011
Exam 1
January 27, 2011

Name: _____

Section: _____

Student ID Number: _____

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- There are 6 questions spanning 4 pages. Make sure your exam contains all these questions.
- You are allowed to use a scientific calculator (**no graphing calculators**) and one **hand-written** 8.5 by 11 inch page of notes.
- You must show your work on all problems. The correct answer with no supporting work may result in no credit. **Put a box around your FINAL ANSWER for each problem and cross out any work that you don't want to be graded.** Give exact answers wherever possible.
- If you need more room, use the backs of the pages and indicate to the grader that you have done so.
- Raise your hand if you have a question.
- Any student found engaging in academic misconduct will receive a score of 0 on this exam.
- You have 80 minutes to complete the exam. Budget your time wisely.
  SPEND NO MORE THAN 15-18 MINUTES PER PAGE!

GOOD LUCK!

1. (12 points) Evaluate the integrals:

(a) $\int x^3 \left(\frac{1}{x^4} - \frac{5}{\sqrt{x^7}} \right) \, dx$

(b) $\int_1^4 \frac{e^{\sqrt{x}}}{\sqrt{x}} \, dx$

(c) $\int \frac{x^3}{(1+x^2)^5} \, dx$

2. (6 pts) If $f(x) = \int_{\sin(5x)}^{2} e^t \sqrt{t+3}\, dt$, find the derivative of $f(x)$ and evaluate it at $x = \pi$. That is, find the value of $f'(\pi)$.

3. (10 pts) A particle is moving on a straight line with acceleration given by $a(t) = 6t$, where t is in seconds. At $t = 2$ seconds, you measure that the velocity of the particle is $v(2) = -15$.

   (a) Find the velocity function, $v(t)$, for the particle at time t.

   (b) Find the **total distance** traveled by the particle from $t = 0$ to $t = 5$.

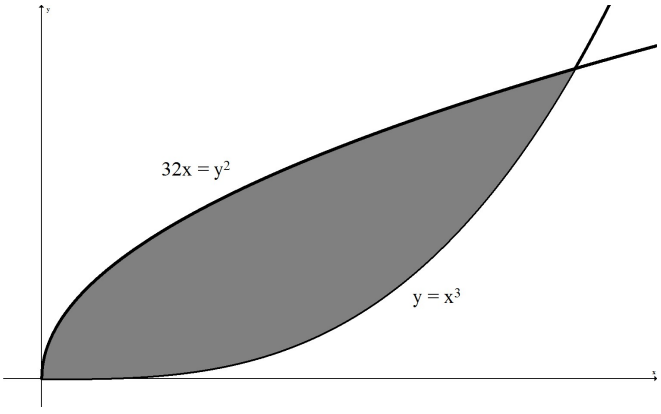
4. Consider

$$\int_1^7 (x^2+1)^{1/3}\,dx$$

(a) (6 pts) Use the left-endpoint rule with $n = 4$ rectangles to approximate the value of this definite integral. Show you work, then give your final answer rounded to 3 digits after the decimal.

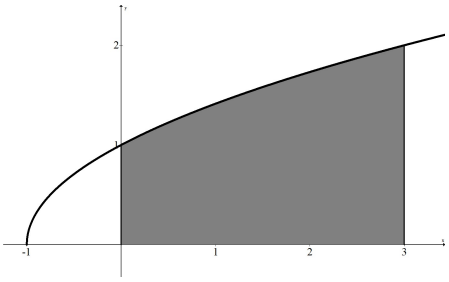
(b) (2 pt) Is your answer an overestimate or underestimate? (You must explain to get full credit)

5. (8 pts) Find the area of the region bounded by $y = x^3$ and $32x = y^2$.



6. (16 points)

Consider the region, R, bounded by the curve $y = \sqrt{x+1}$, the x-axis, and between $x = 0$ and $x = 3$. A picture of this region is given at right.



(a) (4 pts) Set up an integral (DO NOT EVALUATE) for the volume of the solid obtained by rotating the region R about the **horizontal line** $y = -3$.

(b) (6 pts) Find the volume of the solid obtained by rotating the region R about the x-axis. Set up the integral AND evaluate.

(c) (6 pts) Find the volume of the solid obtained by rotating the region R about the y-axis. Set up the integral AND evaluate.